

1. C

2. C

3. A

4. A

5. B

6. A

7. B

8. C

9. A

10. B

11. B

12. C

13. B

14. C

15. B

16. C

17. B

18. C

19. B

20. A

21. supervised learning is a type of machine learning that use the labeled data which means that we have to provide all data even row and column and also expected output. Then the machine will start process all dataset with specific algorithm like Linear Regression or KNN to find the pattern that can reach the the expected output.

25. list out 2 major applications of data science:

- Facebook: facebook has used data science with various of things, but one major thing they use it for analyze user behavior like what user mostly click like on, what video that user like to watch then facebook will start pop out those content to user for recommendation.

~~Blood test in health care. they use it to~~

- Banking: they use it to determine their business plan. After input all data from previous year they can know what is their weakness, what to improve, what to create.

26. Structure data:

- + having table, row, column
- + most data are store in excel file
- + easy to read, clear
- + used for data science
- + Example: excel, csv file

- Semi-structure data:

- + not 100% structure
- + some part clear, some not
- + not easy to read if compare to structure data.
- + Example: xml, json file

27. ~~to find~~ Outlier in data is a process of ~~detect~~ detect the ungroup data. That kind of data can impact result of our machine learning too, we can find it by seeing it in the dataset like their value does not make sense to the dataset. we can determine it by using z-score and cut the outlier data if it is not in our condition.

28. Data preprocessing is a type of process that is used to make sure that our dataset is strong enough to process. It has many steps in data preprocessing such as data probing, ~~normalizing~~, ~~transforming~~.

29. Normalization is a technique that is important in k-NN because it will ~~make~~ make sure that features in the dataset are scaled down between 0 and 1, so the process of k-NN can go better because features seem balanced.

31. ~~K~~ in k-means algorithm, is the amount of objects that we ~~can~~ want the objects to be. In k-means, all objects are not assigned, so we have to give a specific amount of k to ~~ask~~ ask k-means to find the objects.

32. Hyperparameters of machine learning are: dataset, classification metric, normalization, outlier detection.

34. In the below 3 ~~class~~ classifiers, the first one is the best one. I will explain why:

- First one: The boundary line separates the 2 objects well. ^{triangle} ~~circle~~ didn't go to square, the space between boundary line to the dotted line also maximizes. The dotted line to triangle also not touch with each other.
- Second: Both objects have exceeded the boundary line which does not match to SVM concept. Both ~~of~~ objects should not stay in that space.
- Third: Both objects are separated from boundary line well, but we can see that both lines can go to near with ~~to~~ the objects but it didn't. So the first one is the best.

35. BBF work in ~~svm~~ SVM by: trying to maximize the space from boundary line to dotted line and from dotted line to object and make the hyper plane with valley between them, so it ~~easy~~ easier to see.

22. comment M1: it give 99% accuracy for training dataset, and give 30% of accuracy for testing.

for my idea, I think that it has 1/3 different. so at first step we can say, machine didn't use proper algorithm which does not match to their dataset, maybe there is ~~problem~~ with outlier data is testing dataset, missing data in the dataset, did not use classification metric to determine whether it is good or not. So, for my idea, M1 should do data pre-processing, choose proper algorithm, do the data ~~clean~~ cleaning, do the classification metric.

23. Regression analysis is the processing of analyzing the continuous data. it will give result this is on scale. it is used to predict any result after inputting proper data.

24. Machine learning challenge:

- Expensive: because we have to find the ~~data~~ dataset, use resource like computer, human to process. so the cost is also expensive. which is hard for customer to buy.

- accuracy: it will also having problem of accuracy if they don't use proper data, or data is very less or not using proper algorithm and if no testing data it is also the problem.